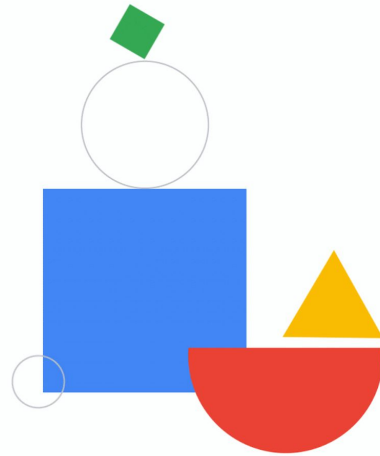


Developing Applications with Google Cloud: Foundations

Module 5: Adding Intelligence to Your Application



Welcome to Developing Applications with Google Cloud: Foundations, module 5: Adding Intelligence to Your Application.

Machine learning is about teaching machines to recognize patterns like humans do. Although even a two-year-old can easily distinguish between an apple and an orange, it's difficult to teach a computer to do the same thing. Google has developed pre-trained machine learning models and made them available to you as easy-to-use Google Cloud APIs. Now, with just a few lines of code, you can add artificial intelligence, or AI, to your own application.

Agenda

01

Using pre-trained machine learning models

02

Introduction to generative AI



In this module, we explore Google's pre-trained machine learning APIs for vision, speech, video intelligence, and natural language processing.

Generative AI takes artificial intelligence a step further. Generative AI is a type of artificial intelligence that creates new content based on what it has learned from existing content. Generative AI can make applications more powerful and the development experience more efficient. This module will introduce you to generative AI, and explain why developers should care about it.

Agenda

01

Using pre-trained machine learning models

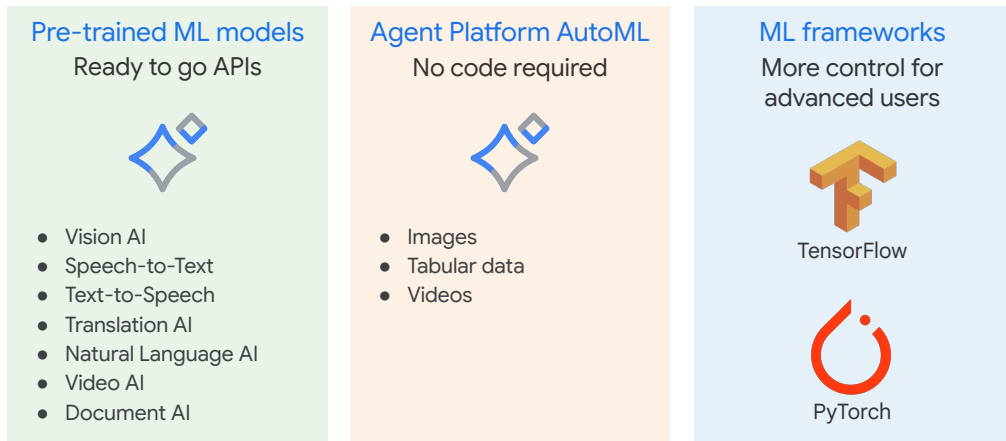
02

Introduction to generative AI



Pre-trained learning models can help you add powerful features to your applications.

Use pre-trained machine learning (ML) models to add intelligence to your applications



Google Cloud offers several pre-trained machine-learning models that you can use to add intelligence to your application.

- Vision AI lets you perform complex image detection.
- Speech-to-Text and Text-to-Speech enable developers to convert audio to text and text to audio.
- Translation AI lets you translate an arbitrary string into any supported language. The Cloud Translation API is highly responsive. Websites and applications can use the Cloud Translation API for fast, dynamic translation of text from a source language to a target language.
- Natural Language AI lets you extract information about entities that are mentioned in text documents, news articles, or blog posts. You can use the Cloud Natural Language API to understand sentiment about your product on social media, or parse intent from customer conversations.
- Video AI lets you search video files to extract and label entities at the shot, frame, or video level. The Video Intelligence API annotates videos stored in Cloud Storage and helps you identify key entities in your video and when they occur within the video.
- Document AI takes unstructured data from documents and transforms it into structured data, making it easier to understand, analyze, and consume.

AutoML on Gemini Enterprise Agent Platform (Agent Platform AutoML) enables users with limited ML expertise to train high-quality models specific to their business needs. Agent Platform AutoML lets you train models on images, tabular data, or videos

without writing any code.

You can also use your own data to build and train your own custom ML models by using frameworks like TensorFlow and PyTorch.

Invoke REST APIs to use machine learning APIs; no machine learning knowledge is required

Invoke Vision API

The Vision API can work off an image in Cloud Storage or embedded directly into a POST message. I'll use C



. That photograph is from <http://www.publicdomainpictures.net/view-image.php?image=15842>

Image (Cloud Storage/
embedded)

JSON request

```
# Running Vision API
import base64
IMAGE="gs://cloud-training-demos/vision/sign2.jpg"
vservice = build('vision', 'v1', developerKey=APIKEY)
request = vservice.images().annotate(body={
    'requests': [{
        'image': {
            'source': {
                'gcs_image_uri': IMAGE
            },
        },
        'features': [{
            'type': 'TEXT_DETECTION',
            'maxResults': 3,
        }]
    }],
})
responses = request.execute(num_retries=3)
print responses

{'responses': [{('textAnnotations': [{('locale': 'zh', 'description': 'u8bf7u6e
```

JSON response

It's really easy to invoke the REST APIs to implement machine learning in your application, no ML knowledge is required. In this example, we are using the Vision API to process an image that's stored in Cloud Storage. We invoke the REST API and send it a JSON request, and we receive a JSON response with attributes that describe the image. Let's take a look at a few examples now.

Analyze images



Label detection



Optical character recognition (OCR)



Landmark detection



Logo detection



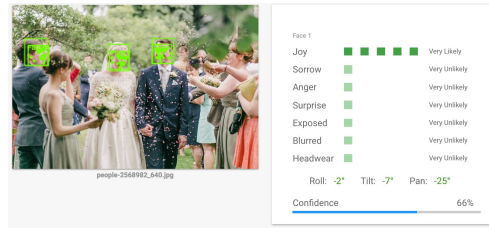
Face detection



Explicit content detection

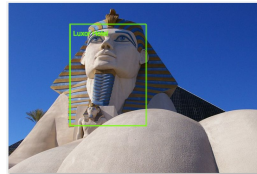
The Vision API can categorize objects under labels and perform optical character recognition, or OCR. The Vision API can detect landmarks, logos, faces, and explicit content.

Get insight from images

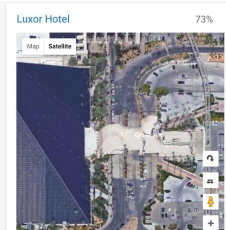


For example, the Vision API can analyze faces and return information about emotions and head wear. In the wedding picture, the API accurately returns the emotional expressions on the faces in the picture.

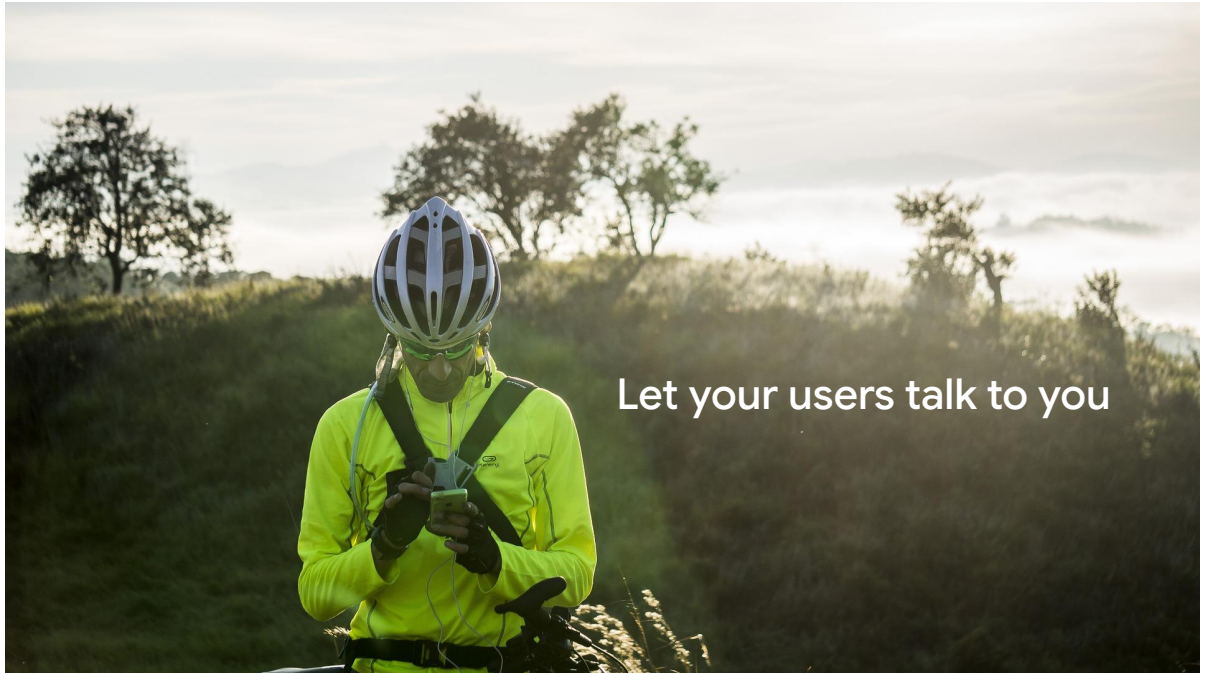
Get insight from images



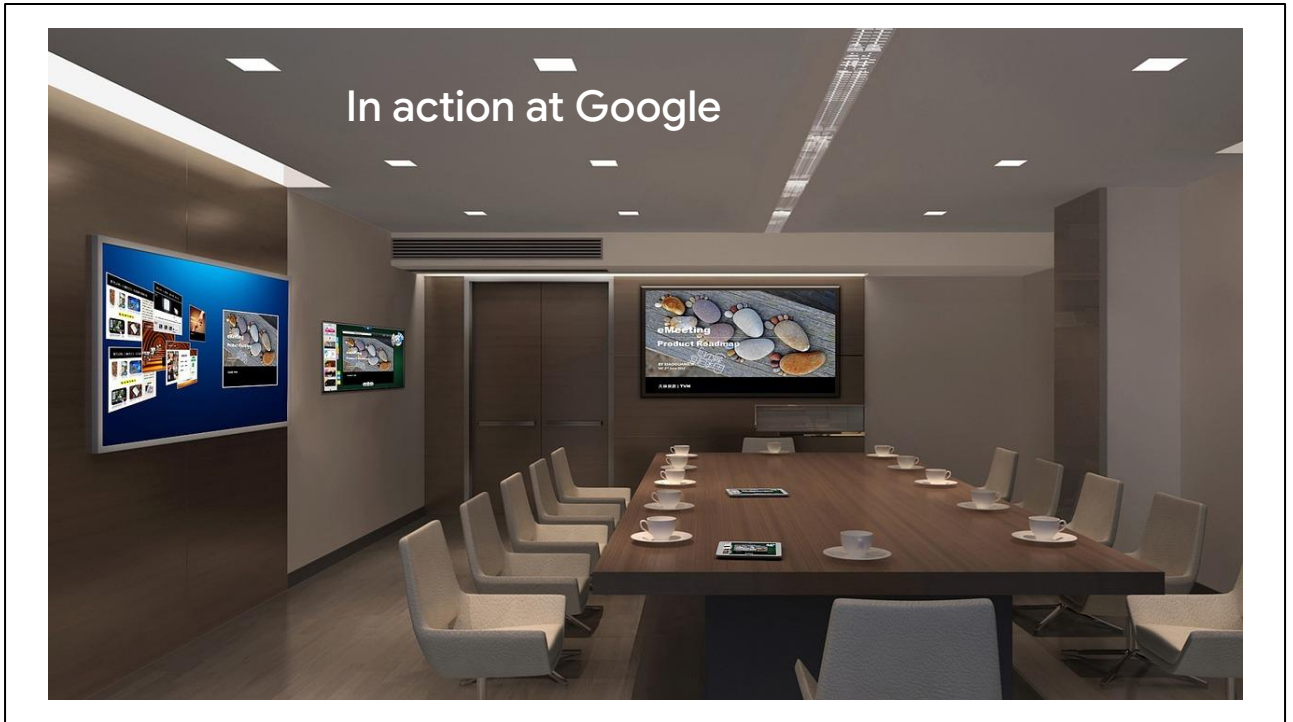
las-vegas-1086414_640.jpg



In the picture of the Sphinx, Vision API correctly detects that the image is from the Sphinx in Las Vegas and not the Sphinx in Egypt.



The Speech-to-Text API enables developers to convert audio to text. It handles 110 languages and variants to support your global user base. You can transcribe the text of users dictating to an application's microphone, enable command-and-control through voice, transcribe audio files, and more.



Here's an example of how Google uses machine learning. Google's conference room systems perform occupancy detection by using motion detection with the VC camera. Every 30 seconds, the system sends a Pub/Sub notification indicating whether motion was detected or not. It also sends a Pub/Sub notification when a call starts or ends.

If motion is detected between 6 and 8 minutes after the meeting start time, the room counts as occupied. Otherwise, it's empty, and is available for someone else to reserve.

Agenda

01 Using pre-trained machine learning models

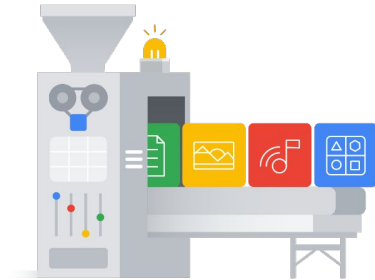
02 [Intro to generative AI](#)



One type of artificial intelligence that you can use for your applications is generative artificial intelligence, or generative AI for short.

What is generative AI?

- Generative AI creates new content based on what it has learned from existing content.
- Learning from existing content ("training") results in the creation of a statistical model.
- When provided a prompt, this model can be used to predict an expected response and generate new content.

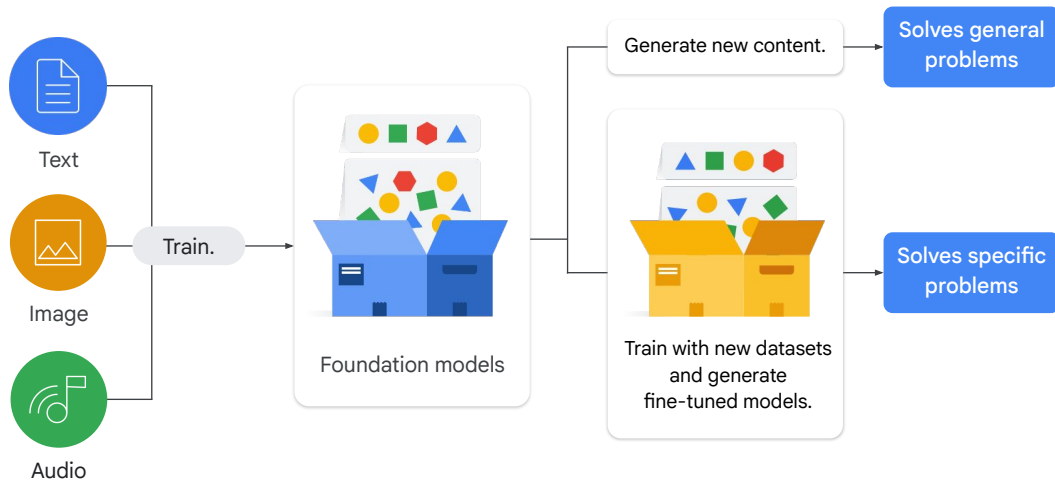


Generative AI is a type of artificial intelligence that creates new content based on what it has learned from existing content.

We call this type of learning "training." A statistical model is created by using the existing content.

You can provide an input, called a prompt, to the model, which can predict an expected response. New content can be generated based on the expected response.

How does generative AI generate new content?



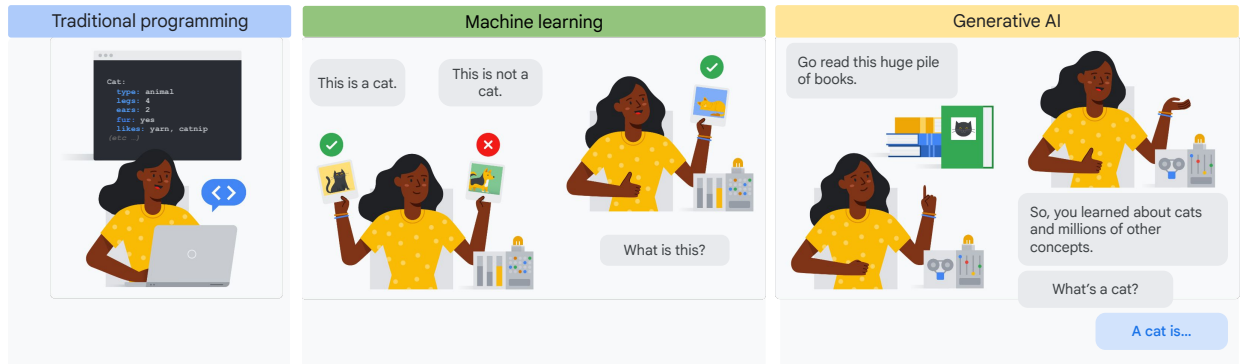
How does AI generate new content? It learns from a massive amount of existing content such as text, images, and audio.

Training results in the creation of a “foundation model.” The most popular type of foundation model is a large language model, or LLM. LLMs are trained on text data only, but other types of foundation models may be trained on other types of data, like images or programming code.

The foundation model can then be used directly to generate content and solve general problems, such as content extraction or document summarization.

The model can also be trained further with new datasets in your field to solve specific problems, such as financial model generation or healthcare consulting. This training results in the creation of a new model that is tailored to your specific needs.

From traditional programming to generative AI



How is generative AI different from traditional programming and other types of machine learning?

In traditional programming, you have to specify the rules, then the machine will act on them and return the answers.

For example, using traditional programming, you might specify these attributes of a cat:

- type: animal
- legs: 4
- ears: 2
- fur: yes
- likes: yarn, catnip

However, writing these algorithms is difficult because it's impossible to implement all possible rules.

So you need a new method: machine learning with neural networks.

With machine learning, you feed the machine data and answers and let it discover the rules itself. For instance, you train the machine on many pictures of cats and other animals. The machine learns the pattern and predicts whether a new picture is a cat.

However, this type of learning is typically in a **narrow** field to solve a **specific** task. What if you want a machine to develop some **fundamental intelligence** to solve

general problems?

Generative AI aims to solve this problem.

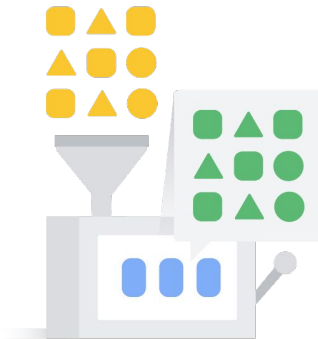
With generative AI, you feed a machine a huge amount of multimodal data. The machine learns a seemingly endless number of concepts and develops **foundation** models like an LLM.

So when you ask the machine “what’s a cat,” it can give you everything it learned about a cat.

Large language models

They are **large**, **general-purpose** language models that can be **pre-trained** and then **fine-tuned** for specific purposes.

- ✓ Large
 - Large training dataset
 - Large number of parameters
- ✓ General-purpose
 - Commonality of human languages
- ✓ Pre-trained and fine-tuned



So, what are large language models? Large language models refer to **large**, **general-purpose** language models that can be **pre-trained** and then **fine-tuned** for specific purposes.

What does large mean? **Large** has two meanings.

First is the enormous size of the **training dataset**, sometimes at the petabyte scale.

Second it refers to the number of **parameters**, which now reaches billions and even trillions. Parameters are essentially the memories and knowledge that the machine has learned during model training. Parameters determine the ability of a model to solve a problem, such as predicting text.

General-purpose means that the models are sufficient to solve common problems. The models work due to the commonality found in a human language, regardless of the specific tasks you are trying to do.

This leads to the last point: **pre-trained and fine-tuned**. A large language model can be pre-trained for general purpose use with a large dataset. Later, it can be fine-tuned for a specific purpose by using a much smaller dataset.

Example generative AI use cases



Create

Use cases

- Generate stories or poems.
- Improve images.



Summarize

Use cases

- Summarize video, audio, and paragraphs.
- Generate Q&A.



Discover

Use cases

- Search for a document.
- Discover products.



Automate

Use cases

- Extract and label contracts.
- Classify feedback and create ticket.

What are the potential use cases of generative AI?

Generative AI can create content and bring your thoughts and visions to life. It can do things like:

- Generate stories or poems based on prompts that you provide, or improve images based on instructions.
- Creation of content is a key benefit of generative AI, but generative AI can do much more.

Generative AI can summarize knowledge. Such as:

- Automatically summarizing video, audio, and paragraphs, or
- generating questions and answers based on the content.

Generative AI can do search and discover for you. For example, it can:

- Search for a document, or
- Discover products based on desired features.

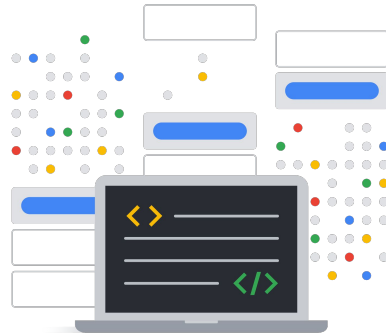
Generative AI can also automate workflows. For instance, it can:

- Extract and label contracts, or
- classify feedback and create support tickets.

You can use generative AI to create powerful and compelling applications.

Generative AI use cases for application development

- Code generation
- Documentation
- Code explanation



- Fixing code
- Code completion
- Code translation

Generative AI revolutionizes how applications are developed.

You code your apps with help from your own generative AI-powered coding assistant.

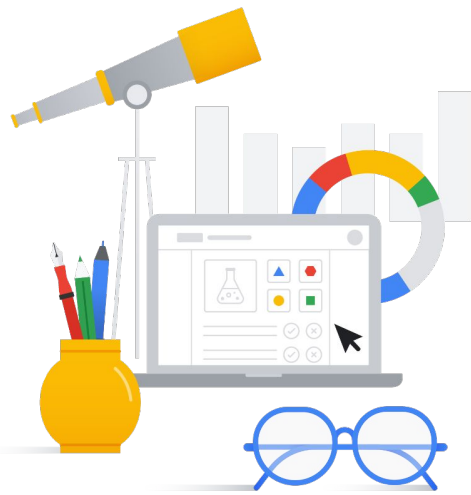
Features include:

- Code generation. You can generate code based on a natural language description of the desired code, and automatically generate unit tests for a piece of code or ask your assistant to optimize code.
- Documentation. The assistant can add comments to your code or generate release notes based on the changes.
- Code explanation. You ask your assistant to explain what the code does, and how it does it.
- Fixing code. Your AI assistant can find bugs in your code, and then fix them.
- Code completion. As you type your code, the context of your code is used to finish the line of code you're writing. Or your code editor might suggest code for the entire function.
- Code translation. Take code written in one coding language, and have your assistant translate it into another, while adhering to coding conventions of the new language.

Models developed by Google assist with code generation, code chat, and code completion to provide these features. Gemini provides you this assistance to help you write code faster and more efficiently.

[Code models overview:

<https://docs.cloud.google.com/gemini-enterprise-agent-platform/models/google-models>]



Hands-on lab:

Adding User
Authentication and
Intelligence to Your
Application

In this lab, "Adding User Authentication and Intelligence to Your Application," you enhance the bookshelf application by using OAuth to authenticate users. You use Secret Manager to store sensitive application data. Finally, you use the Cloud Translation API to translate the description of the book to the preferred language of the logged in user.

In this module, you learned ...



Google Cloud provides pretrained models for vision, speech, video intelligence, and natural language processing.



You can use these pretrained models or train your own models without being a machine learning expert.



Tools like **Agent Platform** make it easy to train and deploy custom machine learning models on Google Cloud.

We began this discussion by looking at the machine learning services, tools, and pretrained models that help you add intelligence to your applications.

Google Cloud provides pretrained models for vision, speech, video intelligence, and natural language processing.

You can use these pretrained models or train your own models without being a machine learning expert. You can then use these models to build exciting features in your applications.

Google Cloud also provides tools like Agent Platform that can help you use your own data to build customized models. Using these models in your apps is as simple as making API calls.

In this module, you learned ...



Generative AI differs from traditional programming.



You learned about large language models.



We discussed Generative AI use cases for a variety of applications and application development tasks.

We also discussed generative AI, and how it differs from traditional programming.

You learned about large language models, and we discussed generative AI use cases for applications and application development.